Step 1: Explore — Data Profile

CAPSTONE-01 | AI-Assisted Audit Control Framework · MCX AI Lab: Data Analysis with AI

Dataset: **grc_risk_register.csv** — synthetic GRC risk register, 160 rows × 19 columns · snapshot date 30 Jun 2026

1 STRUCTURE AND GRAIN

One row per identified risk item. **Item_ID** is a clean primary key — 160 unique values, zero duplicate rows, and the year prefix reconciles to Date_Identified in all 160 records. Coverage runs 04 Jul 2024 to 30 Jun 2026 (24 months); due dates extend to 18 Apr 2027. Nothing in the file post-dates the snapshot.

Identifiers (2)	Item_ID, Control_ID
Dimensions (9)	Department (8), Risk_Domain (9), Risk_Tier (4), Control_Effectiveness (4), Regulatory_Driver (19), Identification_Source (6), Status (6), Owner_Role (8), MAP_Linked (2)
Metrics (4)	Inherent_Risk_Score (3–24), Residual_Risk_Score (1–19), Days_Overdue (0–523), Financial_Exposure_USD ($5K–$7.9M)
Temporal (3)	Date_Identified, Due_Date, Closed_Date
Text (1)	Risk_Title — 43 distinct issue types, reused across departments

2 COMPLETENESS AND INTEGRITY

One column has nulls: Closed_Date, 43 of 160 (26.9%) — all on open-status items, which is structurally correct rather than a defect. Every other column is 100% populated, with no whitespace or case inconsistencies. Referential logic holds throughout: no closed item lacks a close date, no open item carries one, no residual score exceeds its inherent score, and no due or close date precedes identification. **The file is clean. The risk here is not dirt — it is structure.**

3 DATA QUALITY FINDINGS

CRITICAL	**Three columns are derived, not independent.** Risk_Tier is a deterministic banding of Inherent_Risk_Score (Low 3–7, Med 8–13, High 14–19, Crit 20–24, non-overlapping). Financial_Exposure_USD is banded identically. Any finding that “critical risks carry higher exposure” recovers the generator’s rules, not a business insight. Treat all three as one variable.
CRITICAL	**Days_Overdue cannot measure historical timeliness.** 18 closed items were closed after their due date (up to 59 days late), yet all 117 closed items carry Days_Overdue = 0. The field is a point-in-time snapshot for open items only — where it reconciles exactly (43 of 43). Derive on-time delivery from Closed_Date minus Due_Date instead.
CRITICAL	**Status was assigned independent of item age.** Closure rate by identification quarter is flat — 68, 72, 70, 73, 75, 77, 78 and 71 percent across eight quarters. Items identified as little as zero days before the snapshot show the same closure rate as items two years old, which is not possible in a live register. Cohort-over-time closure comparison is therefore invalid. The related on-time curve (85% rising to 100% while median cycle time falls 119 days to 24) is right-censoring, not improvement: in recent cohorts only fast-closing items have had time to appear.
MATERIAL	**17 items closed with untested controls.** 10 sit at Closed - Validated with Control_Effectiveness = “Not Tested”; 7 more at Closed. Validation is being attested without control testing evidence. Most audit-relevant anomaly in the file.
MATERIAL	**Control_ID is 1:1 with Item_ID.** 160 items, 160 unique controls, sequentially matched. No control covers more than one risk. Real registers are many-to-one; this forecloses control-concentration analysis entirely.
WATCH	**MAP linkage carries an unexplained exception.** MAP_Linked = Yes appears only on High and Critical items (zero Low/Medium), reading as a business rule — yet 8 of 15 Critical items are not linked. Either a rule violation or an undocumented threshold.
WATCH	**Exposure is right-skewed; volume is uniform by design.** Exposure skew 2.96 (mean $918K vs median $394K) — report medians and totals, never a mean. Volume holds at 17–24 items per quarter across all 8 quarters: no seasonality, no trend, no change point exists to find.
WATCH	**Status is fragmented.** Open (4), In Progress (3) and Remediation In Progress (3) are ten items across three near-identical labels. Collapse to Open / Past Due / Closed for any visual.

4 ANALYTICAL BOUNDARIES

SUPPORTED BY THIS DATA

- Remediation timeliness and aging — open items average 388 days old (max 720); closed items take a median 106 days
- Control effectiveness mix — 39% Effective, 31% Partially Effective, 17% Not Tested, 9% Ineffective
- Overdue concentration — 33 Past Due items averaging 288 days, max 523
- Recurring issue types across departments (43 titles, some recurring 10× across 5–7 units)
- Model & AI Governance slice — 19 items; 9 EU AI Act, 6 Internal AI Policy, 4 SR 11-7

NOT SUPPORTED

- Any correlation of financial exposure with tier or risk score — the relationship is constructed, not observed
- Any trend, seasonality or change-point claim on risk volume over time
- Control coverage, concentration, or shared-control failure analysis
- Historical on-time closure rates read directly from Days_Overdue
- Cohort-over-time closure or performance comparison — closure rate is flat by construction, and recent cohorts are right-censored

5 CARRY-FORWARD HYPOTHESIS FOR STEP 3 (ANALYZE)

Inherent-to-residual correlation is **0.813**, with a consistent 0.25–0.83 reduction ratio. If residual score proves to be a mechanical haircut off inherent score, then **Control_Effectiveness does not actually drive risk reduction in this dataset** — a finding in its own right, and a sharp target for the adversarial review in Step 4 (Verify).

Source: grc_risk_register.csv (synthetic; no production or employer data). **Method:** Python / pandas profiling — column classification, null and cardinality profiling, cross-column business-rule tests, distribution and correlation checks. All figures computed, none estimated. **Confidence:** 95% on structural findings (directly computed); 75% on intent — distinguishing generator artifacts from modelled business rules requires confirmation from the dataset author. **AI role (transparency):** profiling scripts and draft written by Claude (Opus 5) under human direction; findings reviewed and owned by N. Fragoso. **Prepared:** 27 Jul 2026. **Revised:** 30 Jul 2026 — finding 8 (status assigned independent of item age) added following cohort and censoring tests run during Step 2 framing; analytical boundaries updated accordingly.